Imperial College London

EPSRC Centre for Doctoral Training in NEUROTECHNOLOGY for life and health

Investigating grey matter lesion distribution in Multiple Sclerosis using PET ligand [^{11}C]-PBR28

Sam Podmore, Prof. Paul Matthews

Introduction

[^{11}C]-PBR28 is a widely used second generation radio ligand in **positron emission tomography** (PET) due to its high specificity for the **18 kDa translocator protein** (TSPO). In this study, we used [^{11}C]-PBR28 to assess reactive astrocytes and activated microglia, pathophysiological markers of neuroinflammation in patients with **relapsing-remitting multiple sclerosis** (RRMS) and **secondary progressive multiple sclerosis** (SPMS).

Multiple sclerosis (MS) is a neurodegenerative disease identified by distributed inflammation across the brain, which can localise as lesions. These lesions can be distinguished from **normal appearing white matter** (NAWM) using MRI due to oedema, gliosis and demyelination, however this is not generally the case in the grey matter making **grey matter lesions** (GML) difficult to detect [1].

Here we explore an application of pseudo-referencing in PET imaging to maximise the heterogeneity between GML's and **normal appearing grey matter** (NAGM) in patients with MS.

Aims

- **Identify** reliably clusters of abnormal [^{11}C]-PBR28 uptake that may reflect active, inflammatory grey matter lesions
- **Explore** the impact of modulating pseudo-reference size and properties for identifying such sparsely distributed clusters of high [^{11}C]-PBR28 binding

Methods

In this study 22 Healthy controls and 28 MS patients (22 RRMS / 6 SPMS) underwent [^{11}C]-PBR28 PET with MR imaging [2].

Subjects were separated into two cohorts; **high affinity binding** (HAB) and **mixed affinity binding** (MAB) based on TSPO genotype. Each cohort then underwent the processing pipeline described below in Figure 1.

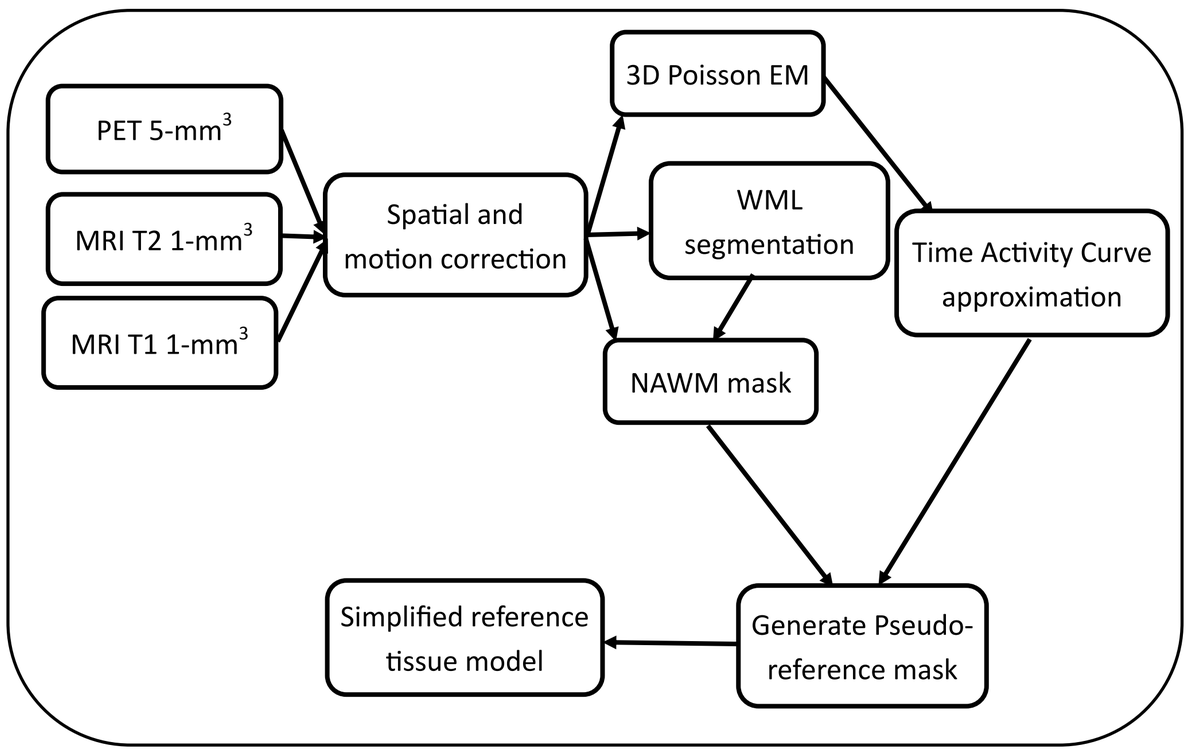


Figure 1. Processing pipeline for PET and MRI

Non-displaceable binding potential maps (BP_{ND}) were generated by **simplified reference tissue model** (SRTM) which describe the amount of ligand that binds specifically (interpreted as marking reactive astrocytes and activated microglia (1).

$$BP_{ND} = \frac{V_T}{V_{ND}} - 1 \qquad (1)$$

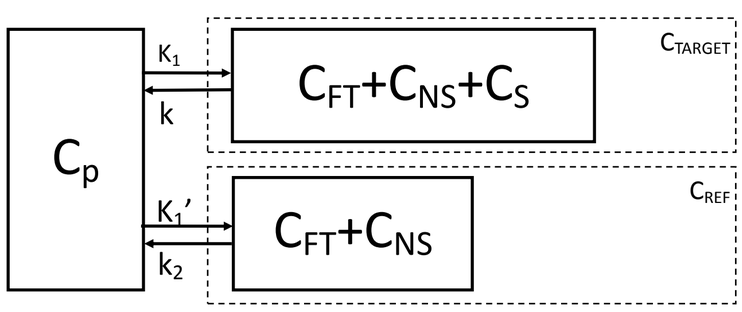


Figure 2. Simplified Reference Tissue Compartment Model

Results

To assess the stability and biological meaningfulness of pseudo-reference regions we generated 30 pseudo-reference regions for each subject based on the mean of healthy controls plus varying amounts of their standard deviation and applied SRTM with each as reference. From Figure 3. we decided to select mean plus half standard deviation for the remainder of the study as it showed asymptotic error reduction in the BP_{ND} and minimised pseudo-reference volume which in turn minimised the risk of selecting voxels with microstructural inflammation (specific binding).

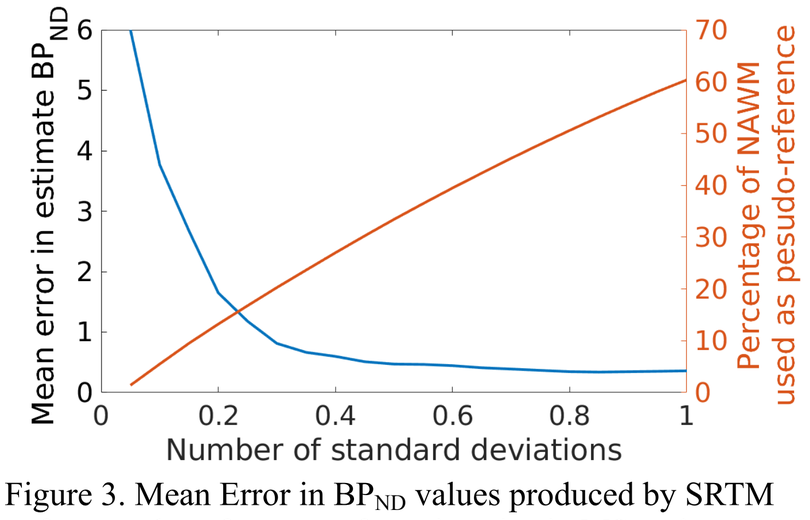


Figure 3. Mean Error in BP_{ND} values produced by SRTM for pseudo-references of varying size in MS patients

Deep grey matter analysis

BP_{ND} maps were segmented based on the CIC atlas and mean BP_{ND} was calculated for each cluster. Significance was calculated using two sample t-test between MS deep GM and healthy controls deep GM adjusted for family wise error rate (FWER, a = 0.05) with Šidák procedure at 12 comparisons. Significant differences were found in the BP_{ND} between MS subjects and healthy controls in the Posterior Parietal ($p < 0.05$), Thalamic Primary sensory ($p < 0.05$) and left and right Predorsal Putamen ($p < 0.05$, $p < 0.05$). Due to the conservative nature of the FWER 4 deep grey regions were close to significance but not considered significant (p<0.08).

Neocortical grey matter analysis

Each subject in the cohorts (HAB & MAB) underwent neocortical segmentation of their BP_{ND} maps. The cohorts then underwent two sample t-testing (group 1. MS, group 2. Healthy control) with FWER (a = 0.05), global mean normalisation and subject to a 10 voxel threshold in SPM12. However No supra-threshold clusters were found to be significant.

Conclusion

Here [^{11}C]-PBR28 has been shown to be successful in detecting neuroinflammation in a portion of the Deep grey matter tissues of multiple sclerosis patients relative to healthy controls however it is limited by many factors. Partial volume effects limit the applicability of traditional statistical tests and the PET spatial resolution inhibits fine grain detection of clusters of neuroinflammation across the neocortex. A definition of an optimal pseudo-reference region was identified here which maximised neocortical heterogeneity. With this and future advancements in post processing it will allow for identification of grey matter lesions in MS pathology in the future.

Acknowledgments

This study was supported by the EPSRC centre for doctoral training in neurotechnology for life and health. The author would like to thank Dr Gourab Datta, for collecting the experimental data and Dr Jim Myers for his technical support throughout the project.

References

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